AI-Powered Hospital Readmission Prevention Program

AI Validation, Bias Testing & Regulatory Strategy Plan Version 1.0
Healthcare AI Implementation Portfolio Project

4.1 Clinical Validation Strategy

The purpose of validation is to confirm that the AI model performs safely and accurately in real healthcare settings.

Internal Validation:
The model is tested using data from the healthcare organization where it was developed.

External Validation:
The model is tested in different healthcare systems to confirm generalizability across different patient populations and workflows.

4.2 AI Performance Evaluation

Model performance will be evaluated using:

Discrimination:
Measures whether the model can separate high-risk and low-risk patients. Metrics include AUROC, precision, and sensitivity.

Calibration:
Determines whether predicted risk matches actual outcomes.

Clinical Impact:
Evaluates whether AI improves patient outcomes, care coordination, and readmission prevention.

4.3 Bias and Fairness Assessment

The model will be evaluated for fairness across demographic groups including race, ethnicity, gender, age, insurance status, and socioeconomic factors.

Important measures include:
- False-negative rates
- False-positive rates
- Accuracy differences across groups

The goal is to ensure the AI solution provides equitable performance and does not disadvantage vulnerable populations.

4.4 Regulatory Strategy: Software as a Medical Device (SaMD)

The AI solution will be evaluated under Software as a Medical Device considerations because it supports healthcare decision-making.

Intended Use:
Predict patients at increased risk of hospital readmission to support care management decisions.

Risk Consideration:

Moderate risk because the system influences clinical workflow but does not independently diagnose or treat patients.

4.5 Human Oversight Plan

The AI system will support clinicians rather than replace clinical judgment.

Workflow:
AI identifies risk -> Clinician reviews information -> Care team determines intervention -> Patient receives appropriate care.

4.6 Post-Deployment Monitoring

After implementation, the organization will continuously monitor:

AI Performance:
- Accuracy changes
- Model drift

Safety:
- Patient outcomes
- Missed high-risk cases

Fairness:
- Performance across demographic groups

Workflow:
- Provider adoption
- Alert fatigue

Document Outcome

This document defines the validation, fairness, regulatory, and monitoring strategy required for safe deployment of an AI-powered hospital readmission prevention solution.